

Automated Building Validator Proposal

Green Metrics Technology

Team Members

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Introduction

Problem Overview

Municipal zoning bylaws define critical rules for land use and development, including building height limits, setbacks, density, and permitted uses. These rules determine whether a proposed building design is feasible on a given parcel of land.

However, across Canada, bylaws are distributed across hundreds of jurisdictions, each with its own document formats, terminology, and update cycles. There is currently no standardized structure or centralized system to interpret these regulations consistently.

As a result, bylaw interpretation remains largely manual, requiring practitioners to search through lengthy documents for each project. This process is time-consuming, error-prone, and difficult to scale. As housing demand increases and permitting processes become more complex, this inefficiency represents a critical bottleneck for development across municipalities.

Motivation and Purpose

Automating the interpretation of municipal bylaws presents a significant opportunity to improve efficiency, consistency, and scalability in development feasibility assessments and decision-making.

Key benefits include:

- Reduced time-to-permit through faster feasibility evaluation
- Consistent and defensible decisions using rule interpretation
- Scalability across jurisdictions by applying a unified framework
- Improved transparency through structured and traceable regulatory data

This project aims to design and prototype a system that transforms fragmented regulatory documents into structured rules that can be applied programmatically.

Background Research

Prior research highlights both the potential and challenges of automated legal document processing.

- **Waddell (2002) [1]** introduces *UrbanSim*, a computational framework for modeling urban development, which relies on structured representations of land use regulations. This work demonstrates the importance of transforming zoning rules into machine-readable formats to enable large-scale analysis and decision-making.
- **Niklaus et al. (2021) [2]** present a multilingual benchmark for legal judgment prediction, showing that NLP techniques can effectively extract and classify information from complex legal documents with conditional language. This is directly relevant to the extraction of structured rules from bylaw text.
- **Axelrod et al. (2023) [3]** explore the use of machine learning and natural language processing to automate zoning data collection across jurisdictions. Their work highlights the challenges of fragmented zoning documents, inconsistent formats, and the difficulty of extracting standardized zoning information at scale. The study also demonstrates the potential of hybrid approaches combining automation with human validation, which is directly relevant to this project's focus on scalable bylaw retrieval and rule extraction.

Together, these studies demonstrate that extracting structured information from legal text is feasible using computational methods. However, a key challenge remains in adapting these approaches to fragmented, heterogeneous, and unstandardized municipal bylaw systems, which motivates the need for a scalable and unified solution.

Data Sources

The municipalities that the solution will support include Vancouver, Richmond, Burnaby, Abbotsford, Coquitlam, Kelowna, Prince George, Kamloops, Hope, and Cranbrook. Each municipality has three relevant data sources: the zoning bylaw PDFs, the parcel geometry file, and the zoning district file. If time constraints become an issue, only a subset of these municipalities will be supported based on data quality.

Zoning Bylaw PDFs

The zoning bylaw PDFs are the primary source for zoning bylaws for a given municipality. Some municipalities have all zoning bylaws contained within a single PDF file, others divide the zoning bylaws into many. Outlined below are several specific geometric bylaws that are required to determine the legality of a building on a parcel based on the building's width and length.

- **Minimum Front Yard Setback:** The minimum distance from the front property line.
- **Minimum Rear Yard Setback:** The minimum distance from the back property line.
- **Minimum Side Yard Setback:** The minimum distance from the side property lines.
- **Maximum Site Coverage:** The maximum percentage of the total lot area that a building can occupy.

Maximum building height will not be considered since it introduces many edge cases such as exempted roof structures. Corner lots will not be supported as well due to their many edge cases in bylaw. Ambiguous bylaw clauses may be encountered, where the system

cannot confidently extract a single rule. If the system detects one of these cases, it will indicate the need for human review instead of giving a possibly incorrect answer.

Parcel Geometry and Zoning District Files

The parcel geometry and zoning district file for a given municipality can usually be found in GIS data formats by searching on the municipality's open data portal. Both files are required for different parcel information. The parcel geometry file provides the actual shape and size of each property. This can be used to calculate the legal buildable area of a given parcel. The zoning district file provides the zoning type of each parcel. The zoning type is needed to determine where to search in the zoning bylaw document for related rules.

Note, the ParcelMap BC [4] is a merged parcel data source for municipalities across BC. However, it is not the authoritative source for legal property boundaries or related record attributes, so its use will be avoided.

Aims and Objectives

Research Questions

This project investigates the following core questions:

- How can bylaw source documents be reliably discovered and retrieved across jurisdictions with no common schema or indexing?
- How can rule-relevant content (definitions, constraints, conditional clauses) be extracted from fragmented, inconsistently formatted bylaw documents?
- How can extracted rules be translated into quantifiable geometric rulesets and clash-detection logic suitable for parcel-level feasibility queries?

Methods and Tools

Methods

The problem will be broken into three independent subproblems: database discovery, rule retrieval, and clash detection. One team member will focus on each of the subproblems. The output of one subproblem can be fed into the input of the next to provide a complete solution.

Database Discovery

There is no single portal to find the data sources. The navigation paths to each data source for each city of interest will be recorded. Then, web scrapers will be built with Python to fetch the needed data. Parcel GIS files can usually be found by searching on a municipality's open data portal. Zoning bylaw documents can usually be found with a simple Google search. The solution should fetch all zoning bylaw documents if there are multiple.

For evaluation, the true data source locations will be obtained for the cities of interest by manually searching. Then, unit tests will be written to determine the success rate for the scrapers to fetch each data source for the cities of interest.

Rule Retrieval

To obtain the zoning type of a given address or PID, the zoning district files will be processed with a GIS solution. Then a RAG solution will be implemented with Python. Different options for RAG architectures will be explored, and costs will be compared. Once an architecture is selected, it will be implemented such that it takes a zoning type and zoning bylaw PDFs as input and outputs the minimum yard setbacks (front, rear, and side) and the maximum site coverage.

For evaluation, the true values for minimum yard setbacks and maximum site coverage will be obtained by manually searching bylaw documents. Then, unit tests will be written to determine the accuracy averaged by municipality and by rule.

Clash Detection

QGIS will be used to build a solution that can determine whether a building is legally permitted on a parcel. The building's width and length will be input. The width and length should be chosen such that it minimizes area while encompassing the building floorplan. If this area is larger than the permitted area determined by the maximum site coverage, then the building is not permitted. Then, the minimum yard setbacks (front, rear, and side) will be used to define offsets from parcel boundaries, resulting in the region that a building can be legally built in. With a representation of the building area and the legal buildable space, translations and rotations can be used to determine if the building fits.

For evaluation, sets of parcels, rules, and building dimensions will be defined with known validity so that unit tests can be written. Varying parcel shapes and sizes will be tested.

Tools

- **Python** — primary language
- **Web scraping** — requests, BeautifulSoup (HTML), pdfplumber (PDFs)
- **AI model** — OpenAI API (GPT-4) or open-source (Mistral, LLaMA) for rule extraction
- **GIS analysis** — QGIS
- **Storage** — GitHub
- **Communication** — Discord
- **Calendar** — Google Calendar

Timeline

This project started April 27 and it must be completed by June 23. The team schedule operates on a weekly basis. There are weekly goals for each team member outlined in Table 1. General comments for weekly team operations are below:

- Each team member will work on this project at least 32 hours a week between Monday and Thursday.
- Each week will start with a one hour meeting at 9:30 AM with the Green Metrics Technology on Microsoft Teams to give a progress update, receive feedback, and discuss the next steps.
- All team members will meet in-person on UBCO campus in EME 1153 from 1-4 PM on Tuesdays for DATA 599 class and information exchange.
- All team members will post a daily check-in on the team Discord when they start work each morning from Tuesday to Thursday. Daily check-ins should include what was done yesterday and what is planned for today.
- All other ad-hoc discussions and meetings related to the project will occur on the team Discord.
- The question and blocker protocol outlined by Green Metrics Technology will be followed by all team members.

Week	Dates	Data Source Discovery (Sasivimol)	Rule Retrieval (Yusen)	Clash Detection (Jamie)
0	April 27 to April 30	- Introduce UBC team to GMT partners - Gain housing development domain knowledge		
1	May 4 to May 7	- Complete project proposal with proposed methods, tools, and timeline		
2	May 11 to May 14	- Start parcel scrapers, get to 50% complete	- Investigate RAG methods - Decide on RAG architecture	- Install QGIS and familiarity with parcel GIS data - Fetch zoning type by address or PID given the needed parcel file (for Rule Retrieval)
3	May 18 to May 21	- Finish parcel scrapers	- Start RAG rule retrieval, get to 25% complete	- Complete solution to define the legal buildable area of a parcel by the geometric rules
4	May 25 to May 28	- Test accuracy for parcel scrapers and refine	- Continue RAG rule retrieval, get to 50% complete	- Start solution to determine whether the building fits in the legal buildable area, get to 33% complete
5	June 1 to June 4	- Start zoning bylaw scraper, get to 50% complete	- Continue RAG rule retrieval, get to 75% complete	- Continue solution to determine whether the legal building fits in the buildable area, get to 67% complete
6	June 8 to June 11	- Finish zoning bylaw scraper	- Finish RAG rule retrieval	- Finish solution to determine whether the building fits in the legal buildable area
7	June 15 to June 18	- Test accuracy for zoning bylaw scraper and refine	- Test accuracy for RAG rule retrieval and refine	- Test accuracy of clash detection and refine
8	June 22 to June 23	- Prepare final presentation - Complete final report		

Table 1: Weekly goals for all team members.

Deliverables

There are three deliverables for this project, with each team member focusing on one, in addition to a final shared report. The output of one deliverable can be fed into the input of the next. Together, they offer a solution to determine whether a building plan of interest is permitted on specific parcels across many BC municipalities. Accuracy metrics will be calculated for each deliverable.

Municipal Bylaw Scrapers

Three Python web scrapers that can take any address or PID as input, and output the related zoning bylaw PDFs, the parcel geometry file, and the zoning district file.

RAG Rule Retriever

A Python RAG solution that takes zoning bylaw PDFs as input and outputs the minimum yard setbacks (front, rear, and side) and the maximum site coverage for a given zoning type. There will also be a GIS solution to determine the zoning type for a given address or PID.

GIS Clash Detector

A GIS solution that can determine whether a planned building can be legally built on a given parcel (identified by address or PID) based on the building's width and length, along with the minimum yard setbacks (front, rear, and side) and the maximum site coverage.

Conclusion

This project aims to address the difficulty of interpreting fragmented municipal zoning bylaws across multiple jurisdictions in British Columbia. Current processes are manual, time-consuming, and difficult to scale. To solve this problem, the project proposes a pipeline that combines automated bylaw retrieval, RAG-based rule extraction, and GIS-based feasibility checking. The system transforms unstructured zoning documents into structured rules that can be used to evaluate whether a proposed building can legally fit on a parcel. If time allows, end-to-end provenance will be implemented.

References

- [1] - Waddell, P. (2002). UrbanSim: Modeling urban development for land use, transportation, and environmental planning. *Journal of the American Planning Association*, 68(3), 297–314. <https://www.tandfonline.com/doi/abs/10.1080/01944360208976274>
- [2] - Niklaus, J., Chalkidis, I., & Stürmer, M. (2021). *Swiss-Judgment-Prediction: A multilingual legal judgment prediction benchmark*. Proceedings of the Natural Legal Language Processing Workshop 2021, 19–35. <https://aclanthology.org/2021.nllp-1.3/>

[3] - Axelrod, J. (2023). *The potential of machine learning for compiling standardized zoning data*. Urban Institute.
<https://datacatalog.urban.org/the-potential-of-machine-learning-for-compiling-standardized-zoning-data-c0f4697a9f0>

[4] - Land Title and Survey Authority of British Columbia. (2026). ParcelMap BC parcel fabric. B.C. Data Catalogue. <https://catalogue.data.gov.bc.ca/dataset/parcelmap-bc-parcel-fabric>